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PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> javac week14/Q1.java
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q1
1(a) Runnable sum = 5050
1(a) Time = 203300 ns

1(b) Thread sum = 5050
1(b) Time = 283000 ns
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> javac week14/Q2.java
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q2
2(a)/(b) Synchronized with join:
Final counter = 10000
Time = 2239600 ns

2(c) Synchronized without join:
Final counter may be less than 10000 when printed immediately = 6000
Time = 779200 ns

2(d) Not synchronized:
Final counter less than 10000 = 8903
Time = 999000 ns

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PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> javac week14/Q3.java
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q3 10
n = 10
Normal sum = 25501
Normal time(in nanosecond) = 600 ns
Normal time(in millisecond) = 6.0E-4 ms
Thread sum = 25501
Thread time(in nanosecond) = 419600 ns
Thread time(in millisecond) = 0.4196 ms
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q3 1000
n = 1000
Normal sum = 3295913
Normal time(in nanosecond) = 17500 ns
Normal time(in millisecond) = 0.0175 ms
Thread sum = 3295913
Thread time(in nanosecond) = 385200 ns
Thread time(in millisecond) = 0.3852 ms
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q3 100000
n = 100000
Normal sum = 337501082
Normal time(in nanosecond) = 251200 ns
Normal time(in millisecond) = 0.2512 ms
Thread sum = 337501082
Thread time(in nanosecond) = 2182400 ns
Thread time(in millisecond) = 2.1824 ms

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PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q3 10000000
n = 10000000
Normal sum = 33844426416
Normal time(in nanosecond) = 7464600 ns
Normal time(in millisecond) = 7.4646 ms
Thread sum = 33844426416
Thread time(in nanosecond) = 9490500 ns
Thread time(in millisecond) = 9.4905 ms
PS C:\Users\abhij\OneDrive\Documents\CODING\CPS393\CODE\SQL\src> java week14.Q3 1000000000
n = 1000000000
Normal sum = 3383580633525
Normal time(in nanosecond) = 738172600 ns
Normal time(in millisecond) = 738.1726 ms
Thread sum = 3383580633525
Thread time(in nanosecond) = 224158600 ns
Thread time(in millisecond) = 224.1586 ms

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